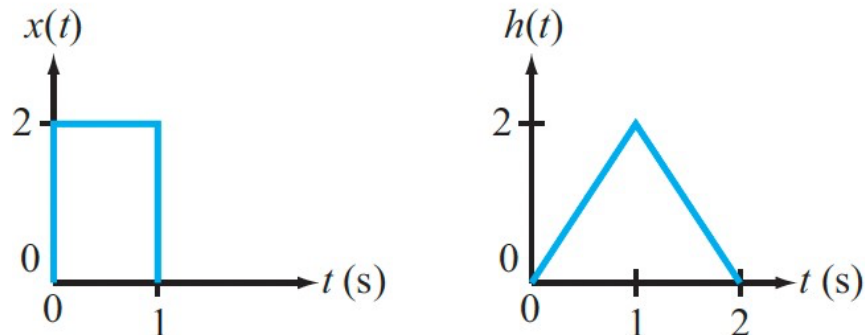


1. Functions $x(t)$ and $h(t)$ have the waveforms shown below. Calculate and plot

$y(t) = x(t) * h(t)$ by:

- (20 pts) performing the convolution analytically, and
- (20 pts) performing the convolution graphically.



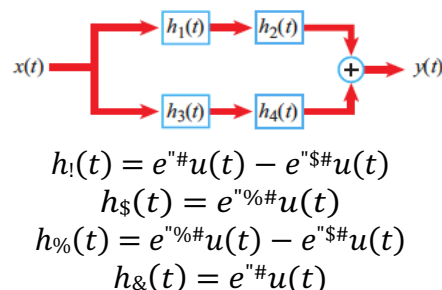
2. (20 pts) Calculate the following convolution *without computing any integrals*, but rather, by using the properties of the convolution integral (see Table 2-1). Cite each of the properties from table 2-1 that you used in your answer.

- $u(t) * [\delta(t+2) - u(t+1) + u(t-1) + 2\delta(t-2)]$
- $u(t) * [(t-1)u(t-1)]$

3. (20 pts) Compute the following convolutions:

- $e^{-2t}u(t) * e^{-3t}u(t)$
- $e^{-3t}u(t) * e^{-3t}u(t)$

4. (20 pts) Calculate the overall impulse response of the following interconnected system:



5. (0 points) Prepare for the next class by reviewing convolutions, and attempting the convolutions Table 2.2. A helpful YouTube lecture is also available for you: <https://youtu.be/xAW7g3iGjf8>. The next worksheet will require you to derive each of these convolutions on your own. Integration by parts is an essential skill for this activity.